

Market Microstructure Frameworks for Spoofing Detection: Isolating Manipulative Intent from Legitimate Liquidity Provision

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Abstract

In this paper we detect the bad guys and we find how to tell them they are actually bad guys. Hopefully, we will be able to do this without accusing the good guys of being bad guys.

Keywords. Spoofing detection; market manipulation; limit order book; high-frequency trading; anomaly detection; market maker; market microstructure

1 Introduction

The transition of global financial exchanges to fully electronic, continuous double auction matching engines has fundamentally transformed the mechanics of price discovery and liquidity provision. In modern financial ecosystems, the Limit Order Book (LOB) serves as the central nervous system of the market, aggregating the continuous stream of bid and ask limit orders submitted by diverse market participants. While the advent of algorithmic and high-frequency trading (HFT) has undeniably narrowed bid-ask spreads and enhanced short-term market efficiency, it has concurrently introduced profound vulnerabilities. The microsecond latency environment allows sophisticated algorithms to exploit the structural mechanics of the LOB, executing complex manipulative strategies that deceive heuristic-based trading algorithms and human traders alike. Among these manipulative practices, spoofing represents one of the most pervasive and challenging threats to market integrity. Spoofing is a deceptive algorithmic strategy wherein a market participant places a massive, non-bona fide order on one side of the LOB with the explicit intent of canceling it before execution. The presence of this immense, artificial volume creates a false impression of directional market pressure. This artificial pressure manipulates the micro-price and induces other market participants—particularly algorithms that rely on order book imbalance indicators—to adjust their quotes or submit market orders. Once the price has been artificially driven to a favorable level, the spoofer executes a smaller, bona fide

order on the opposite side of the book and immediately withdraws the massive deceptive orders, thereby profiting from the artificially induced price dislocation. The European Securities and Markets Authority (ESMA), through the Market Abuse Regulation (MAR) [3], strictly prohibits such practices, categorizing them as behaviors that disrupt the functioning of the trading system and create misleading signals regarding the supply and demand of financial instruments. For regulatory and supervisory authorities such as the Commissione Nazionale per le Società e la Borsa (CONSOB), the detection and prosecution of spoofing present a severe analytical challenge. The physical actions comprising a spoofing strategy—posting limit orders, updating quotes, and rapidly canceling orders—are mechanically indistinguishable from the legitimate actions of a bona fide market maker providing liquidity. Both spoofers and market makers exhibit exceedingly high order-to-trade ratios (OTTR) and operate at high frequencies. Traditional market surveillance mechanisms, which historically relied on static heuristic thresholds, volumetric alerts, or simple cancellation ratios, are fundamentally inadequate in this environment. These legacy systems generate an overwhelming volume of false positives, frequently flagging legitimate liquidity provision and inventory management as manipulative market abuse.

2 The Microstructural Divergence: Market Making Versus Spoofing

To construct a mathematical model capable of detecting market manipulation, it is first necessary to rigorously define the divergent objective functions that govern legitimate market making and spoofing. While both participants generate massive volumes of limit order messages and cancellations, their underlying economic motivations, risk profiles, and optimizations are fundamentally opposed.

2.1 The Market Maker’s Problem

A market maker functions as a continuous liquidity provider, simultaneously quoting resting limit orders at the best bid and the best ask. The primary objective function of a market maker is to maximize the utility of gains derived from capturing the bid-ask spread across a vast number of transactions, while concurrently minimizing the variance of their portfolio and strictly managing inventory risk. In market microstructure theory, the pricing dynamics of a liquidity provider are governed by the interplay between the arrival rate of market orders and the accumulation of inventory.

The optimization problem for a market maker is frequently modeled using stochastic control techniques and Hamilton-Jacobi-Bellman (HJB) equations, where the dealer sets optimal quotes to balance the asymmetric arrival of buy and sell orders. ¹ If a market maker accumulates a long position (positive inventory), they are exposed to the risk of the underlying asset depreciating. To mitigate this inventory risk, the rational market maker will lower their quotes downwards—lowering their ask price to incentivize incoming buyers, and lowering their bid price to dissuade incoming sellers. ² The bid and ask prices are dynamically adjusted functions of the mid-price, the base spread required for profitability, and an inventory penalty parameter that increases as the dealer’s position deviates from neutrality.

Consequently, a market maker’s high cancellation rate is entirely reactive and protective. Limit orders are rapidly canceled and re-posted to reflect minute changes in the fundamental mid-price, to defend against adverse selection informed traders, and to recalibrate inventory penalty. The massive volume of orders and cancellations produced by a market maker is therefore characterized by symmetry across the LOB and a continuous presence at the top of the book to execute trades. Their expected directional gain from posting an order is near zero, as their profit is entirely derived from the spread rather than from inducing a directional shift in the mid-price.

2.2 The Spoofer’s Problem

In stark contrast, a spoofing agent does not seek to capture the bid-ask spread, nor do they operate as symmetric liquidity provider. A spoofer’s objective function is purely directional. The spoofer desires to execute a relatively small, bona fide order at a highly favorable price by artificially manipulating the micro-price dynamics of the LOB. To execute this strategy, the spoofer deploys a massive non-bona fide order on the opposite side of the LOB. This deliberate injection of spurious volume creates an extreme unbalanced environment, artificially inflating the visible depth of the book. In modern electronic markets, countless algorithmic traders utilize strategies that weight the volume at the bid against the volume at the ask to predict short-term price distributions. By overwhelming one side of the book, the spoofer pollutes these volume imbalance metrics, causing HBL algorithms to mathematically deduce an impending price shift and aggressively alter their own quoting behavior.

The critical mathematical dilemma for the spoofer lies in the placement of the non-bona fide order. The spoofer must balance the expected market manipulation gain against the explicit risk of unintended execution. If the spoofer places the massive non-bona fide order exactly at the top of the book (the best bid or ask), they generate the maximum psychological and algorithmic impact on the micro-price. However, doing so exposes the spoofer to massive adverse selection and inventory risk; an incoming aggressive market order could inadvertently execute against the spoofer’s massive fake order, leaving the spoofer holding an enormous, unwanted position. To optimize this expected cost function, sophisticated spoofers rarely post their deceptive orders at the absolute best price. Instead, they optimize the posting distance (measured in ticks or basis points) deeper within the book. By placing the order at Level 2, Level 3, or beyond, the spoofer ensures the volume is still visible to the matching engine’s data feed—and thus factored into the volume imbalance metrics of victim algorithms—while maintaining a structural buffer against immediate execution. Once the intended directional shift occurs and the spoofer’s small bona fide order is executed, the massive non-bona fide order is instantly canceled. The spoofer’s cancellation intent is not to manage inventory, but to terminate the manipulative episode the exact millisecond the cost optimization concludes.

3 Existing Approaches

The literature on spoofing detection is still relatively small, but it has moved from transparent microstructure rules toward sequence and graph models that try to capture richer strategic

Table 1: Core Differentiators for Surveillance Models

Microstructure Metric	Legitimate Market Maker	Manipulative Spoofing Agent
Primary Objective Function	Maximize bid–ask spread capture while minimizing inventory variance.	Minimize execution cost of a directional trade via artificial micro-price impact.
Order Book Imbalance	Generally symmetric, adapting quotes to maintain a neutral inventory.	Extremely asymmetric, deliberately creating massive, sudden imbalances (unbalanced quoting).
Cancellation Motivation	Reactive: mitigating adverse selection and updating quotes to track fair value.	Proactive: terminating the deceptive posture immediately after a targeted execution.
Posting Distance Optimization	Top of the book (Level 1) to maximize the probability of spread execution.	Deeper in the book (Level 2/3) to maximize visibility while minimizing execution risk.
Expected Directional Gain	Zero or negative (willing to pay the spread to lay off excessive inventory risk).	Strictly positive (the massive order statistically subsidizes the execution of the small order).

behavior. For the purpose of this paper, the most useful distinction is not simply between “statistical” and “machine-learning” models. It is between methods that identify a stylized spoofing episode from order-book mechanics, methods that monitor abnormal order-flow states, and methods that learn temporal or relational patterns from proprietary event data. Figure 1 summarizes this taxonomy.

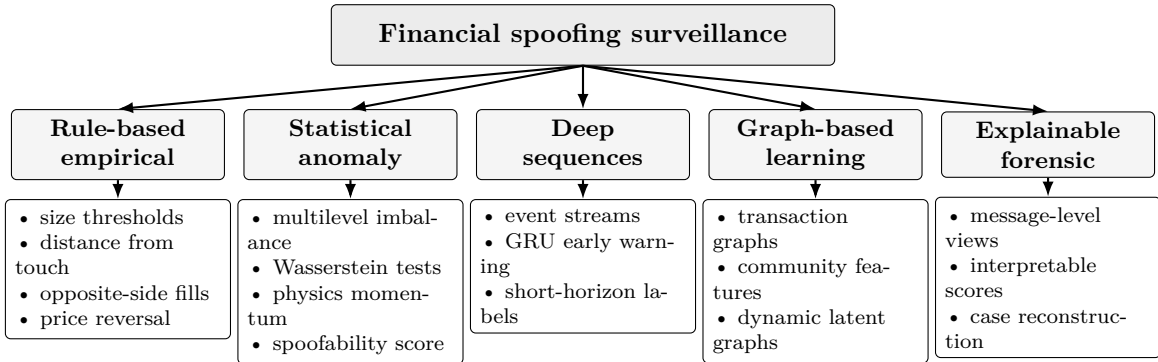


Figure 1: Method families in the spoofing-detection literature. The categories are not mutually exclusive, but they highlight the main modeling choice made by each strand of work.

Early empirical papers define spoofing through economically interpretable order-book conditions. The seminal study by [7] identifies large orders placed far from the best quote, followed by opposite-side trading and subsequent withdrawal, using account-level data from the Korea Exchange. Similar surveillance logic appears in Brazilian equities [9], FX spot markets [11],

cross-market FX settings [10], and Taiwan derivatives [1]. These studies are valuable because the alert logic can be explained to supervisors: suspicious activity is tied to order size, posting distance, cancellation timing, opposite-side execution, and short-lived price impact. Their main limitation is that thresholds are venue-specific and often depend on privileged message data or account identifiers.

A second line develops statistical and anomaly-oriented monitoring. [12] derive spoofing incentives from multilevel order-book imbalance and propose Wasserstein-distance monitoring for real-time detection. [8] use a statistical-physics representation of the order book to detect abnormal momentum in crypto markets. More recent explainable approaches also fall partly in this family: forensic visualization reconstructs message-level spoofing episodes in prosecuted cases [2], while interpretable probabilistic neural networks estimate the conditional gain from manipulating a centralized crypto order book [4]. These methods improve on pure threshold rules by modeling the state of the book, but many papers still report limited benchmark metrics or rely on non-public datasets.

Machine-learning work mainly adds temporal or relational structure. [13] formulate spoofing as an early-warning problem and train a GRU model on cryptocurrency order-book event streams. Graph-based methods then move the unit of analysis from isolated orders to networks of traders and transactions, which is important for detecting coordinated or “conspiracy” spoofing [5, 6, 14]. The strongest reported graph models combine temporal transaction graphs, community-level features, and dynamic latent representations, but their practical value is difficult to assess without open labels and chronological cross-market validation.

Overall, the literature suggests two requirements for a robust surveillance model. First, the features should remain microstructure-grounded: a detector should know whether suspicious volume is large, distant from the touch, rapidly cancelled, and conditional on an opposite-side execution. Second, the evaluation should respect the enforcement setting: labels are noisy, true positives are rare, and false positives impose substantial analyst costs. This motivates the agent-level, spatio-temporal framework developed below, which keeps the interpretability of rule-based surveillance while using the full order-flow sequence available in proprietary data.

4 Data

5 Model

In this section we provide the base reasoning and metrics used to distinguish between legitimate market making behaviour and possible spoofing strategies.

5.1 A Spatio-Temporal Framework for Spoofing Detection

To rigorously distinguish manipulative spoofing from legitimate liquidity provision, we define the microstructural signature of the strategy as a conditional deformation of the agent’s limit-order-book footprint. The relevant object is not only the size of a single order at the best bid or ask, but the whole agent-specific liquidity profile over the first n price levels. This is important because a sophisticated spoofer may place the non-bona fide order away from the absolute best



Figure 2: Schematic top- n spoofing timeline. The large non-bona fide order is represented as an agent-specific depth profile over several bid levels, while the bona fide order sits close to the opposite best quote. The same construction applies symmetrically to sell- and buy-intent episodes.

quote, for example at Level 2, Level 3, or across several adjacent levels, in order to remain visible while reducing the probability of unintended execution.

We formalize the execution timeline of a spoofing agent i into three states:

1. **State 1 (Deceptive Posturing):** shortly before the execution, the agent posts a large non-bona fide liquidity profile on one side of the book, typically away from the touch, while simultaneously maintaining a smaller bona fide order close to the opposite best quote.
2. **State 2 (Execution):** the artificial depth profile distorts short-horizon imbalance signals and the bona fide order is executed.
3. **State 3 (Risk Termination):** immediately after the execution, the agent cancels the opposite-side liquidity profile, thereby terminating the execution risk associated with the deceptive orders.

The model below generalizes the best-level construction to the top n levels while preserving the same economic logic: a large asymmetric profile must be posted shortly before the fill, must be present before the fill, and must disappear conditionally after the fill. This timing restriction is important: long-lived resting liquidity that happens to be on the opposite side of a later execution should not be mechanically interpreted as part of a spoofing episode.

5.1.1 Top- n Agent-Specific Depth Profiles

Let $s \in \{b, a\}$ denote the bid or ask side of the book, and let $k = 1, \dots, n$ denote the book level, with $k = 1$ corresponding to the same-side best quote. For agent i , define

$$V_k^{i,s}(t) \tag{1}$$

as the volume posted by the agent at time t on side s and level k . We denote by

$$\mathbf{V}_n^{i,s}(t) = (V_1^{i,s}(t), \dots, V_n^{i,s}(t)) \tag{2}$$

the corresponding top- n agent-level depth vector. In applications where the book depth varies substantially across assets or intraday periods, the raw volume can be normalized by the total

visible market depth at the same side and level:

$$\tilde{V}_k^{i,s}(t) = \frac{V_k^{i,s}(t)}{D_k^s(t)}, \quad (3)$$

where $D_k^s(t)$ is the total displayed market depth on side s and level k . The distance of a level from the touch is denoted by $d_k^s(t)$. In a tick-time perspective, $d_k^s(t)$ can be measured as the number of ticks between level k and the same-side best quote, shifted by one unit so that the top level has positive distance. This avoids assigning zero weight to Level 1 while preserving the ranking of deeper levels.

5.1.2 Multilevel Distance-Weighted Imbalance

The first ingredient is a scalar projection of the top- n depth profile. We assign to each level a depth kernel

$$\omega_k^s(t) = \frac{e^{-\lambda d_k^s(t)} (1 - e^{-\kappa d_k^s(t)})}{\sum_{\ell=1}^n e^{-\lambda d_\ell^s(t)} (1 - e^{-\kappa d_\ell^s(t)})}, \quad (4)$$

where $\kappa > 0$ controls the increase in execution-risk protection with distance from the touch, while $\lambda > 0$ controls the decay of informational visibility at deeper levels. The product form captures the trade-off faced by a spoofer: a deceptive order should be sufficiently far from the touch to reduce execution risk, but not so far that it becomes irrelevant for short-horizon imbalance signals.

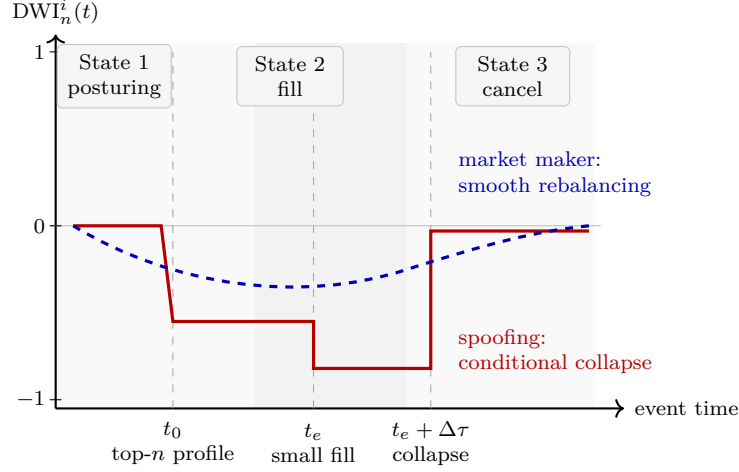
The agent's weighted top- n liquidity on side s is then

$$L_n^{i,s}(t) = \sum_{k=1}^n \omega_k^s(t) \tilde{V}_k^{i,s}(t). \quad (5)$$

We define the *Multilevel Distance-Weighted Imbalance* for agent i as

$$\text{DWI}_n^i(t) = \frac{L_n^{i,a}(t) - L_n^{i,b}(t)}{L_n^{i,a}(t) + L_n^{i,b}(t)}. \quad (6)$$

With this sign convention, $\text{DWI}_n^i(t) > 0$ indicates an ask-heavy agent profile, while $\text{DWI}_n^i(t) < 0$ indicates a bid-heavy agent profile. Because the denominator normalizes by the total weighted agent footprint, the statistic is comparable across agents and trading periods. The case $n = 1$ recovers a best-level version of the measure, whereas larger n values allow the detector to capture deceptive liquidity placed behind the touch.



For the opposite spoofing direction, the sign pattern is mirrored.

Figure 3: Stylized event-time behavior of the multilevel distance-weighted imbalance. A spoofing episode produces a large top- n imbalance and, after the small opposite-side execution, a rapid return toward zero when the deceptive profile is cancelled. Inventory rebalancing by a market maker should evolve more smoothly.

5.1.3 Conditionality and Intent Isolation

A static multilevel imbalance is not sufficient to isolate manipulative intent. A legitimate market maker may also exhibit temporary imbalance while managing inventory or reacting to adverse selection. The discriminating information is the conditionality of the subsequent cancellation: for a spoofer, the disappearance of the large liquidity profile is mechanically tied to the execution of the smaller bona fide order on the opposite side.

Let e denote an execution of agent i at timestamp t_e , and let $s_e \in \{b, a\}$ be the side of the executed bona fide order. The suspected deceptive side is the opposite side, denoted by $\bar{s}_e$. We define

$$t_e^- = t_e - \epsilon_t, \quad t_e^+ = t_e + \Delta\tau, \quad (7)$$

where ϵ_t is chosen so that t_e^- represents the book state immediately before the execution, while $\Delta\tau$ bounds the mechanical latency of the cancellation response. We also impose a pre-execution episode window $\Theta > 0$. An opposite-side order is considered part of the candidate deceptive profile for execution e only if its first observed posting time τ_o satisfies

$$0 \leq t_e - \tau_o \leq \Theta. \quad (8)$$

In the empirical implementation below we use $\Theta = 10$ minutes as an exploratory default. The candidate deceptive orders used for episode attribution are therefore restricted to same-client, opposite-side top- n orders posted within this pre-execution window. The unrestricted top- n state remains useful for describing the agent's book footprint and for computing DWI and side-collapse statistics, but the intent-isolation step should not attribute old standing liquidity to the spoofing episode.

The multilevel analogue of the Spoofing Conditionality Index is

$$\text{SCI}_{e,n}^i = \left| \text{DWI}_n^i(t_e^-) - \text{DWI}_n^i(t_e^+) \right|. \quad (9)$$

However, the top- n representation allows us to be more selective than a pure imbalance difference. We can directly measure how much of the weighted liquidity on each side disappears after the fill:

$$C_{e,n}^{i,s} = \frac{[L_n^{i,s}(t_e^-) - L_n^{i,s}(t_e^+)]_+}{L_n^{i,s}(t_e^-) + \varepsilon}, \quad s \in \{b, a\}, \quad (10)$$

where $[x]_+ = \max\{x, 0\}$. A spoofing episode should produce a large value of $C_{e,n}^{i,\bar{s}_e}$, because the liquidity that vanishes is expected to be on the side opposite to the bona fide execution. By contrast, a general market-maker quote refresh induced by a mid-price change should tend to affect both sides more symmetrically.

We therefore define the *Multilevel Spoofing Conditionality Index* as

$$\text{MSCI}_{e,n}^i = \text{SCI}_{e,n}^i C_{e,n}^{i,\bar{s}_e} \left[C_{e,n}^{i,\bar{s}_e} - C_{e,n}^{i,s_e} \right]_+. \quad (11)$$

This score is high only when three conditions hold simultaneously: the agent's top- n imbalance changes abruptly, the collapse occurs on the side opposite to the execution, and the opposite-side collapse is stronger than the same-side collapse. This construction preserves the interpretability of the original SCI while reducing false positives generated by ordinary two-sided quote updates.

5.1.4 Multilevel Conditional Probability Score

To aggregate repeated behavior over a session, let N_i be the number of relevant small-order executions by agent i . For a fixed depth n , we define the *Multilevel Conditional Probability Score* as

$$\text{MCPS}_n^i = \frac{1}{N_i} \sum_{e=1}^{N_i} \mathbb{I}(\text{MSCI}_{e,n}^i > \gamma_n), \quad (12)$$

where γ_n is an empirically calibrated threshold for the selected book depth. This is the direct multilevel analogue of the original CPS: it counts the fraction of executions after which the agent exhibits a sufficiently large conditional, opposite-side collapse of its top- n liquidity profile.

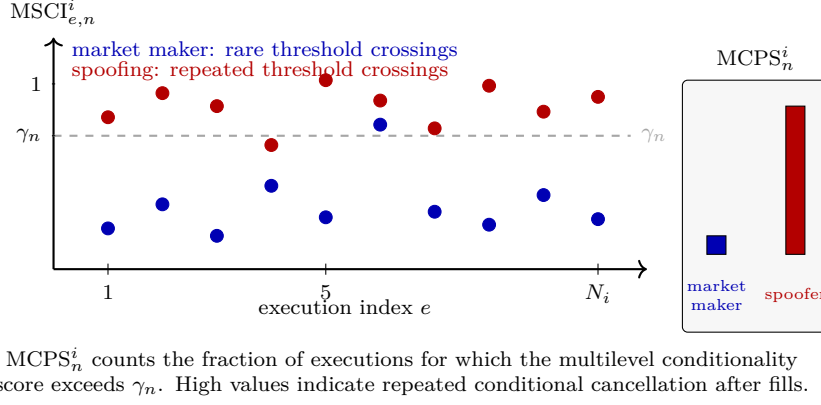


Figure 4: Operational interpretation of the Multilevel Conditional Probability Score. Each point is one execution for the agent. The score increases when the multilevel conditionality index repeatedly exceeds the threshold γ_n .

In practice, the depth n should be chosen according to the venue, tick size, and typical liquidity distribution of the instrument. A robust surveillance implementation can compute MCPS on a grid $\mathcal{N} = \{1, 2, 3, 5, 10\}$ and retain the resulting vector

$$\left(\text{MCPS}_1^i, \text{MCPS}_2^i, \text{MCPS}_3^i, \text{MCPS}_5^i, \text{MCPS}_{10}^i \right) \quad (13)$$

as an interpretable multiscale alert profile. A spoofer operating close to the touch should be detected at small n , whereas a strategy that layers deceptive volume deeper in the book should produce larger values only once deeper levels are included.

5.1.5 Event Reconstruction

The above quantities should be computed from order-level data rather than from aggregate snapshots whenever possible. The reason is that a resting order may move from one level to another because the best quote changes, even if the agent has not cancelled or modified the order. For intent isolation, the relevant event is the removal of agent-owned liquidity after the execution of the bona fide order. The operational procedure is therefore:

1. reconstruct the active orders of agent i immediately before t_e ;
2. assign each active order to its side s and level $k \leq n$, retaining as candidate deceptive orders only those on the opposite side whose first observed posting time lies in $[t_e - \Theta, t_e]$;
3. compute $L_n^{i,b}(t_e^-)$, $L_n^{i,a}(t_e^-)$, and $\text{DWI}_n^i(t_e^-)$;
4. update the book through the interval $[t_e, t_e + \Delta\tau]$ using executions, cancellations, and amendments;
5. compute the corresponding post-event quantities at t_e^+ ;
6. evaluate $\text{MSCI}_{e,n}^i$ and aggregate it through MCPS_n^i .

This order-level reconstruction prevents ordinary repricing of the book from being mechanically interpreted as cancellation of deceptive liquidity, thereby reducing false positives for legitimate market makers.

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